

# Mouloud MAKHLOUF

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## Experience

September 2025 – December 2025

### ■ National Frequency Agency (ANFR) : Research engineer

- **Queonda** : Designed and implemented an LLM-based agent using Retrieval-Augmented Generation (RAG) and GraphRAG techniques to support ANFR agents in answering complex questions, improving information retrieval, and automating high-effort operational processes.

October 2023 – August 2025

### ■ National Frequency Agency (ANFR) : Apprentice at the Data-Lab

- **Propex** : Implemented predictive models and used clustering methods to forecast a meteorological phenomenon called abnormal propagation, which impacts proper TNT reception.
- **Eaprs** : Built a database of cadastral parcels for specific institutions (hospitals, schools, and daycares), used to grant authorizations for radio-electrical site installations.

January 2022 – April 2023

### ■ Private Tutor (via GoStudent platform)

Taught C programming and algorithms to first-year computer science undergraduates.

## Education

January 2026 – now

### ■ Université Paris Cité

PhD candidate on using neuro-symbolic approaches for enhancing LLM-based planning agents

September 2023 – September 2025

### ■ Université Paris Cité

Master's in Machine Learning for Data Science (MLSD)

- M2 MLSD 2024-2025 (GPA 15.90/20)
- M1 MLSD 2023-2024 GPA 16.82/20

Courses : *Natural Language Processing (NLP), Machine Learning, Big Data, Algorithm Complexity, Combinatorial Optimization, Data Science, Clustering, Mixture Models, Reinforcement Learning, Time Series, Deep Learning, Recommender Systems.*

## Education (suite)

September 2021 – June 2023

### ■ Université Paris Cité

Bachelor's in Computer Science

- L3 informatique 2022-2023 **GPA 15.51/20**
- L2 Informatique 2021-2022 **GPA 15.33/20**

Courses : *Graph Theory, Automata Theory, System Programming, Logic, Object-Oriented Programming, Functional Programming, Databases.*

September 2019 – June 2021

### ■ Higher National School of Computer Science, ESI Algiers

Integrated Preparatory Classes

Courses : *Calculus 1,2,3,4, Algebra 1,2,3, Electricity, Electronics 1-,2, Machine Architecture, Probability.*

## Projects

### ■ University Research Project on Using LLMs for Semi-Supervised Learning

Using large language models (LLMs) to enhance clustering outcomes by generating constraints and improving input data quality.

### ■ ENS Data Challenge (Prediction of Future Traffic at SNCF Stations in Île-de-France region)

Applied classical models, LSTM networks and neural networks for medium-to-long-term traffic forecasting.

### ■ XPAT Solver (Ocaml)

Card game solver that returns a solution if a game is solvable, leveraging graph theory and logic.

## Skills

Languages

- French : Bilingual (C1)
- English : Professional working proficiency (B2)

Development Tools

- Python, R, C, Git, pandas, Matplotlib, TensorFlow, dplyr, Scikit-learn, NumPy, Docker, Shiny, MLflow, Spark, Apache Airflow, PyTorch, Flutter, Java, Ocaml, LaTeX,  $\LaTeX$

Databases

- SQLite, PostgreSQL, MongoDB, Cassandra, GraphQL

## Certifications and Awards

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### 📖 Certifications

- DALF (C1) – Advanced French Language Diploma
- TOEFL (B2, Score : 89/120)
- Coursera Machine Learning Specialization
- Coursera R Programming
- Kaggle Introduction to machine learning & pandas library
- Python for everybody (Coursera)
- Google IT Support Specialist (5/6) (Coursera)
- Civil Security Citizen Skills Certificate (PSC1)

### 📖 Distinctions

- **4th** Place, Francilian Programming Day (JFP) 2023

## Interests

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- 📖 Telescope assembly workshops
- 📖 Astronomy and Astrophysics.
- 📖 Reading